Sentiment Analysis For Financial News and Social Media Using ML Techniques

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Abstract: This review article explores the changing dynamics of sentiment analysis in the financial sector, discussing its methodologies and applications, and challenges. The literature underlines how classical lexicon-based methodologies have now given way to techniques using machine learning and, more especially, transformer networks. Merging sentiment analysis with financial data yields important insights into market trends, stock forecasts, and risk evaluation. It discusses the advantages and disadvantages of each methodology, outlines the paramount importance of domain-specific lexicons, and the importance of addressing such challenges as sarcasm, negation, and context-sensitive sentiment interpretations. At the end of the review, it does indicate very strong potential for sentiment analysis in the financial domain, given the recent developments that finally enable hybrid models to capture the strengths of lexicon-based approaches and the robustness of machine learning methods. Directions of further research are also provided, including the development of more robust and interpretable models.

Keywords: Sentiment analysis, Finance; Lexicon-based approaches; Machine learning; Transformer models ; Stock market prediction; Risk assessment; Natural language processing (NLP); Financial forecasting; Sentiment indicators

I. Introduction :

Increased involvement of digital media in finance replaces the dimensions wherein market trends and forecasting were made. Of all, sentiment analysis has been acting as a crucial link in the financial decision-making process. Integrating this with machine learning techniques helps in increasing the predictive power of the financial models, especially. This paper focuses on the fast evolution of sentiment analysis in finance, from its traditional roots in lexicon-based approaches to state-of-the-art machine learning models featuring systems like transformers and hybrids.

This research indicates that this sentiment analysis is actually indispensable in conducting news articles, social media posts, and financial reports to forecast the stock market, manage risks, and make investment decisions.

It emphasizes domain-specific lexicons, discusses some of the problems caused by the interpretation of contextual sentiment, and points out that research into more robust and interpretable models is ongoing. The paper goes through existing methodologies, from machine learning to NLP; it hence underlines the current advantages and disadvantages in the domain and points to possible future directions of research. Coupled with financial data, the sentiment analysis thus provides quite useful insight into how market behavior could be influenced by mass sentiments, where psychological and emotional factors are of critical importance in finance.

II. Literature Review

1. Financial Sentiment Lexicon Analysis[1]

The literature review helps highlight how sentiment analysis has been evolving, more so in financial domains, all the way back since the contribution of Bollen et al., who used OpinionFinder and GPOMS as tools against Twitter data to show that with an accuracy of close to 87%, one is able to predict the stock market by analyzing sentiment. Loughran and McDonald followed up on this by creating a word list related to finance in particular, based on SEC filings, to thereby increase the relevance of approaches based on lexicons in financial sentiment analysis. It further elaborates on applications of machine learning, such as Wang et al., where SVM, Naive Bayes, and Decision Trees were applied on StockTwits tweets labeled as either "bullish" or "bearish" with SVM accuracy of 76.2%. StockTwits is a platform centered on financial discourse and forms a very useful data repository in conducting studies that involve sentiments of a crowd. The comparative study that was held between the lexicon-based approach—VADER, SentiWordNet, and TextBlob—and models developed using machine learning found that

VADER outperformed other methodologies in sentiment extraction from StockTwits posts due to its practicality and precision. The research places a highlight on the role of sentiment analysis in financial forecasting and calls for further research into the continuous evolution of lexicon-based methods for the achievement of higher accuracy in market predictions from social media sentiment.

2. A Survey On Sentiment Analysis Methods, Applications, And Challenges[2]

This literature survey indicates a wider exploration of sentiment analysis, stating that it is significant across domains such as business, healthcare, and social media, among others, where it analyzes the data that users generate in order to give an idea of what people think about and thence any decision. The authors classify sentiment analysis methodologies as machine learning, lexiconbased approaches, and their hybrids, pointing out the respective advantages and disadvantages of all the methodologies. This paper has mentioned NB and SVM as the accuracy and conventional methods for supervised machine learning techniques in the field of Sentiment Classification. Further, it describes how sentiment analysis can be used, such as in product reviews and market research. It also indicates some of the challenges in sentiment analysis such as informal language, sarcasm, domain-specific terminology, in broader challenges resulting from the complexity of human emotions, in order to get the sentiment interpretation accurate. The survey makes suggestions that future research in this area should expand the analysis comparatively on methodologies and face these persistent challenges; sentiment analysis is still a relatively underdeveloped study issue.

3. LSTM-Based Sentiment Analysis For Stock Price Forecast[3]

The liter Of primary importance would be the strengthening of the deep learning methodologies, with emphasis on LSTM neural networks, for forecasting and recognition tasks, such as the prediction of stock prices and text sentiment analysis. By and large, one of the most advanced methods in achieving advanced forecasts of stock prices would be the integration of sentiment analysis through the use of data from news articles and online forums, such as PTT, along with historical stock transaction data. The BERT model is pivotal since it classifies the sentiment of textual material, whether it is positive or negative, hence making sure that it is high in precision. Data from different sources is required for adequate analysis. However, the research has 12.05 %the improvement in forecasting accuracy use of RMSE. The study suggests that sentiment analysis may in fact bring major reductions in error to stock forecasting models, evidence to the prominence of emotions in market dynamics, and proposes future research for nuancing emotional classifications for more precise predictions.

4. Sentiment Analysis In Finance: From Transformers Back To Explainable Lexicons (Xlex)[4]

The literature survey traces the evolution of sentiment analysis in finance but with a focus on lexicon-based approaches such as that of Loughran-McDonald, which is quite effective but proves to be rather time-consuming in its development and maintenance due to manual annotation. The domain specificity of the lexicons is highlighted; while some words mean different sentiment polarities in finance compared with other fields, this sets the ground for the necessity of specialized vocabularies. This has led to the proposition of statistical methods for the automatic improvement of lexicons through social media data, while more recently, the emergence of deep learning models, like transformers, has transferred interest to more powerful though less interpretable methods. In order to balance performance and explainability, hybrid approaches like the one lately suggested in the case of XLex are here to be put in place, combining the strengths of both lexicon-based methods with transformer models, and SHapley Additive exPlanations. The XLex methodology provides better coverage of vocabulary and classification accuracy compared to the LM lexicon, while at the same time being more interpretable. These points give relevance to very tightly scoped financial sentiment analysis. Experimental results confirm the effectiveness of the proposed XLex method, with very strong improvements in accuracy over traditional sentiment classification lexicons. It concludes the survey on the potential of explainable lexicons to move forward sentiment analysis in finance and other domains, possibly followed by future research into their robustness and how to integrate them into other NLP techniques.

5. Evaluation Of Transformer Models For Financial Targeted Sentiment Analysis In Spanish[5]

The paper provides a deep review of the literature in sentiment analysis (SA) available for the financial domain from Spanish texts. It emphasizes the evolution of techniques in NLP, with special emphasis on the role and effect of LLMs like BERT, RoBERTa, ALBERT in carrying out the sentiment analysis. It also brought out the problem

of complexity in the financial language, which is contextdependent, hard to interpret, and classifies the sentiment correctly. The survey underlines the need to target sentiment analysis, in which sentiment is especially analyzed with respect to the main economic target in financial texts. The sentiment may vary according to the context and the stakeholders involved, and it therefore requires that these targets are exactly identified. Several research efforts applying NLP techniques to Spanish texts are reviewed by authors, noticing the development of specialized datasets for better sentiment classification accuracy. Comparing the performances of several models, this paper concludes that deep learning approaches have, in general, surpassed others, especially when fine-tuned for Spanish; therefore, it has portrayed better results in classifying sentiment. The performance went well on classifying sentiment toward the MET, society, or other companies for models like MarIA and BETO. The paper also discusses the use of multi-label classification models, but such an approach, at best, proves only comparable to the single target model. It is finally suggested that the most promising avenue may be through ensemble learning, combining multiple LLMs, but it has still proved quite challenging to avoid the worst predictions that degrade overall performance. Future research directions include extending the dataset to improve model performance, exploring other NLP frameworks like Stanza, and investigating the relationship between financial data and political ideology. This survey lays a solid foundation for the paper's exploration of transformer models in financial sentiment analysis, especially within the Spanish-speaking context.

6. A Comparison Of Lexiconbased And Mlbased Sentiment Analysis: Are There Outlier Words?[6]

Paper "Comparing Lexical-Based and ML-Based Perceptual Analysis: Are There Distinctive Features?" It provides a comprehensive comparison of dictionary machine learning (ML) approaches for sensitivity analysis. It highlights the strengths and weaknesses of each approach, with syntax-based approaches relying on pre-defined text sense criteria, while ML-based approaches, such as Microsoft Azure Text Analytics, evolve towards it specific areas by training Study hedonometers

(based on dictionaries) and analyzes sensory data from over 150,000 texts in four areas using Azure (ML-based) to explore possible differences in between the two methods Regression analysis showed that the importance of words varied by region, but neither was systematically different in perceptual scores between the two methods No consistent outlier word was found Findings showed that although dictionary-based methods may be effective, they may not always be consistent with MLbased ones, emphasizing the need for careful consideration when choosing tools for sensitivity Analysis Future research could examine the effects of specific words in perceptual analysis and in order to do so examine other models for better accuracy.

7. A Comparative Study Of Lexicon-Based And Machine Learningbased Classifications In Sentiment Analysis[7]

The paper "A Comparative Study of Dictionary-Based and Machine Learning-Based Classification in Sentiment Analysis" reviews the most common methods used in sentiment analysis. It compares dictionary-based methods that use predefined words to track emotions with machine learning methods. These methods classify objects using known patterns. Machine learning can overcome context-sensitive dictionary methods. SVM, Naïve Bayes, and Random Forests provide better accuracy by helping to optimize datasets. Studies show that machine learning outperforms pure cellular methods in terms of reliability. Hybridization methods are also discussed here. These combine multiple algorithms with aspect-based full sentiment analysis, which requires special raw materials. The study notes the need for sensitivity analysis as online searches grow in popularity. It emphasizes the continuous improvement of these methods to better capture and capture consumer sentiment.

8. Text Analysis In Finance: The Challenges For Efficient Application[8]

Klimczak 2020, "Economic Analysis: Challenges in Implementation Efficiency" offers a survey of the current state of development in regard to text analysis in economics. The covered topics include various computational linguistics techniques applied to a large variety of languages and domains, such as sentiment analysis, topic detection, segmentation, and languages.

Literature review The literature review focuses the research on the critical financial literature, putting forward the idea that while accounting is vastly studied, the role of context in the decision-making process is often not considered. It also reviews methods of topic recognition, with special emphasis placed on latent Dirichlet distribution, and finally pays attention to the key issues of performance, which are hardly ever noticed in the economics literature.

It highlights challenges for textual analysis, including systematic investigations into the language of finance, formulation of corporate capabilities, consideration of specific strategies, and perceptions of production and communication context about. Communication concludes this chapter by calling for further research into the understanding of economic challenges.

9. Comprehensive Review Of Text-Mining Applications In Finance[9]

It does indicate, in this study, the role of text mining in finance and financial applications, such as in financial forecasting, banking, and corporate finance. This is of great importance to the research paper. In addition, it is currently a listing of research considerations into determining when there are opportunities for future growth in the growing field. The insights presented herein are useful with the growing impact of text mining in the economy, characterized by a potential for further innovation.

10. Behaviour And Markets: The Interaction Between Sentiment Analysis And Ethical Values[10]

The paper "Behaviour and Markets: The Interaction Between Sentiment Analysis and Ethical Values?" deals with the problem of how sentiment analysis interacts with ethical language within financial markets. More precisely, the authors delve into the role of emotions and moral language in explaining financial scandals. In this paper, they present an analysis of the Enron case with respect to negative and vice-related terms derived from the General Inquirer dictionary, looking at their relationship with market sentiment and Enron's stock returns. The results obtained are somewhat complex in their implications regarding the interplay between moral language and financial performance. However, it is noted that this study encounters problems in measuring this effect precisely due to the limitations of the GI dictionary. Additional research is called for, specifically an increase of the dictionary and further case studies, in order to appreciate better how ethical language can affect markets.

11. Sentiment Analysis Of Financial News: Mechanics And Statistics[11]

The paper "Sentiment Analysis of Financial News: Mechanics and Statistics" is focused on exploring the predictive power of sentiment indicators based on financial news, more precisely at the prediction of equity prices, volatility, and trading volumes. While the authors do agree that sentiment data might not always directly show expected causality with market outcomes, they point out that by themselves, they have important predictive capabilities. This further elaborates on the rigor of the methodologies of techniques to construct sentiment indicators and, therefore, their statistical properties. Practical guidelines are then provided for financial practitioners, calling for more research in order to go beyond the existing limitations and thus improve techniques for sentiment analysis within finance.

12. Sentiment Analysis In Financial Texts[12]

The paper introduces a new sentiment analysis engine for the analysis of financial text that extends the conventional methods through linguistic analysis at both the word and the phrase levels. The research will investigate how sentiment from unstructured text data, for instance financial news and reports, affects the market trend and will give a tool to the investors for the assessment of sentiments in huge financial text streams. This SAE with a lot of techniques, such as machine learning and parsing, is more superior compared to the traditional bag-of-words models for sentiment detection, showing itself rather effective in handling large volumes of textual data. These assessments indicate that the method generates useful information, reflecting market dynamics through collective sentiment expressed in various textual forms and, therefore, providing concrete assistance for proper decision implementation by financial practitioners.

13. Sentiment Analysis Using Domain-Adaptation And Sentence-Based Analysis[13]

The paper "Sentiment Analysis Using Domain-Adaptation and Sentence-Based Analysis" covers the progress made in sentiment analysis, more specifically, a transition from word-level to sentence-level approaches. Herein, it proposes a method that enables the adaptation of general-purpose polarity lexicons towards specific domains, together with new sentence-level features such as sentence length, purity, and subjectivity, in order to further enhance sentiment classification. The research principally emphasizes the impact of sentence-level analysis on sentiment classification accuracy, taking the test case of hotel reviews from TripAdvisor. Though two-class sentiment classification has turned out to be solid, multimodal class classification marks a challenge for the paper and points to further tasks in research with regard to sentence-based features and integrating word embeddings for improved performance.

14 . Domain Adaptive Learning For Multi Realm Sentiment Classification On Big Data – PMC[14]

The paper focuses on advanced sentiment analysis by employing transfer learning methods and domain adaptation techniques, hence tending to the issues of big data and domain variability. The traditional methods that are enumerated in the paper include the Support Vector Machine (SVM) and Naive Bayes. Also highlighted are the challenges posed when dealing with such kinds of data.
The work is driven by advanced transformer models like BERT, RoBERTa, and ELECTRA, tuned by the FARM framework of sentiment classification in problems with multiple domains, under the realization of strict labeled data limitations. Empirical results are demonstrated on a number of datasets containing hotel reviews and tweets, whereas transfer learning is proven to advance performance dramatically in sentiment analysis compared to other traditional machine learning models, such as CNN and LSTM. The paper underlines that these methods are poised to be applicable in the future to other domain adaptation tasks.

15. Sentiment Analysis For Effective Stock Market Prediction[15]

This paper presents a new attempt at stock market prediction by fusing sentiment analysis from Really Simple Syndication news feeds with sensex points, which are very similar to stock market index data. The given study will try to reveal that sentiments expressed in feeds have a major influence on the fluctuation of stock markets. The predictive model has been presented by the authors with data collections and analyses of RSS feeds and stock market data for an ARBK company listed on the Amman Stock Exchange. The model showed better results compared to traditional algorithms like ID3, C4.5, and moving average indicators by 14.43%. The paper is oriented toward integrating sentiment analysis with technical indicators, such as guiding investments in the stock market, and in the future it is planned to include social media content to increase the quality of the predictions.

16. Stock Trend Prediction Using Sentiment Analysis – PMC[16]

This paper explores the integration of sentiment analysis with stock trend prediction by introducing a weighted sentiment index that combines Twitter retweet counts, VADER sentiment scores, and FinBERT-generated news sentiment. It emphasizes the importance of timing by proposing an
"opening hours division" method to analyze sentiment data and predict next-day stock movements. The study also incorporates the effects of holidays in its sentiment scoring. Various machine learning algorithms, including KNN, SVM, random forest, and Naive Bayes, were evaluated, with Naive Bayes showing the most effective performance. The paper concludes that combining sentiment analysis with a focus on timing and external factors can significantly enhance stock trend prediction accuracy.

17. Impact Of Social Media Sentiments In Stock Market Predictions: A Bibliometric Analysis[17]

The paper "Impact of Social Media Sentiments in Stock Market Predictions: A Bibliometric Analysis" presents a detailed bibliometric analysis with respect to research into stock market prediction using social media sentiment during the period from 2010 to 2020. In this approach, 1,450 documents have been downloaded from the Scopus database to identify top journals, influential authors, frequently cited papers, and common research keywords within the domain. Sentiment analysis from these platforms, including Twitter, has been an area of growing interest to be incorporated into the prediction models of the stock market. The paper points out the United States as the leading contributor in this area of research and cites future opportunities for research, such as its application in other databases and languages.

18. Sentiment Analysis Of Financial News Using Unsupervised Approach[18]

This paper considers improvements in sentiment analysis techniques for the important financial domain, sensitivity in which heavily affects the decision-making process. Building on Turney's unsupervised sentiment classification method again, but introduced in his seminal 2002 paper, the study applies this to financial news. Evaluation of two alternative techniques—a noun-verb combination and a hybrid model—confirms they both outperform traditional methods in capturing sentiment from financial text, with the noun-verb approach the best performing. The work contributes to improvements in sentiment analysis frameworks, where the limits of working techniques are highlighted and more effective models are proposed in the financial context.

19. A Mass Psychological Perspective On Financial Markets[19]

The paper "A Mass Psychological Perspective on Financial Markets" criticizes the traditional economic view for developing models that do not include psychological and social factors as the drivers of market behavior.

Traditional theories, premised on rational expectations and utility maximization, it holds, do not allow for the explanation of such phenomena as asset price bubbles or market overreactions. While it speaks for a mass psychological perspective, it underlines the role of collective sentiment and social dynamics in financial markets. Recent financial crises are classified by the paper as examples of first and second order mass phenomena and call for empirical research to validate these concepts. Moreover, policymakers have to implement psychological insight into market regulation and seek a sustainable turn in order to reduce future crises.

20. Challenges In Sentiment Analysis[20]

The paper "Challenges in Sentiment Analysis" gives a limitation of the current approach in Sentiment Analysis methodologies and key areas for improvement. Major challenges identified include the fact that there is a need for advanced techniques to handle complexities such as negation and sentiment composition, the context in which sentiment is interpreted needs to be understood, and the task of accurately annotating the sentiment data. The paper also points out the challenges in sentiment analysis across languages and calls for design choices to be target application-based. It, in general, calls for further research into the development of more fine-grained, context-aware sentiment analysis systems that can handle such challenges.

Sr. no	Title	Author	Method used	Advantages	Limitations	Results
1	Financial Sentiment Lexicon Analysis	Sahar Sohangir, Nicholas Petty, Dingding Wang	Lexicon-based sentiment analysis Comparison with machine learning models	Lexicon-based methods are fast and efficient. VADER outperforms machine learning for financial sentiment.	Only 10% of StockTwits messages labeled. Machine learning needs large training data	Sentiment analysis enhances stock price prediction. 75% accuracy using top authors' sentiment.
2	A Survey On Sentiment Analysis Methods, Applications, And Challenges	Annavarapu Chandra Sekhara Rao, Mayur Wankhade	Supervised Learning Unsupervised Learning Semi-supervised Learning Lexicon-Based Approaches Hybrid Models	Supervised learning is highly accurate and effective; unsupervised learning needs less effort and data; semi-supervised learning excels in ambiguity; lexicon-based methods offer structured sentiment classification.	Supervised learning requires labeled data, significant effort. Unsupervised learning has unproven performance. Semi-supervised learning is sensitive to noisy data. Lexicon-based approaches lack depth in emotion capture.	Comprehensive evaluation of sentiment analysis techniques. Identified advantages and disadvantages of methods. Highlighted hybrid models' importance for better performance.
3	LSTM-Based Sentiment Analysis For Stock Price Forecast	Ching-Ru Ko , Hsien-Tsung Chang	LSTM Neural Networks BERT	Improved forecast accuracy by combining sentiment analysis with	Simplified sentiment classification lacks nuanced emotion recognition.	RMSE improved 12.05% using LSTM with sentiment data.

				data. State-of-the-art NLP tools adapt to context. Handles multiple data sources effectively.	Data specificity limits model's generalizability across markets.	
4	Sentiment Analysis In Finance: From Transformers Back To Explainable Lexicons (Xlex)	Maryan Rizinski, Hristijan Peshov, Kostadin Mishev, Milos Jovanovik, Dimitar Trajanov	Transformer Models (FinBERT, RoBERTa) with SHAP	Combines lexicon-based and transformer methods for accuracy. SHAP ensures interpretability of sentiment scores. Automates lexicon maintenance, reducing manual effort.	High computational cost for training transformers. Not ideal for real-time sentiment analysis.	XLex model improves vocabulary coverage significantly. XLex+LM lexicon achieves 0.450 accuracy.
5	Evaluation Of Transformer Models For Financial Targeted Sentiment Analysis In Spanish	Ronghao Pan, José Antonio García-Díaz, Francisco Garcia-Sanchez, Rafael Valencia-García	Transformer Models Supervised Deep Learning	High accuracy with MarIA and BETO, achieving 76.04% F1-score. Versatile in analyzing multiple targets and handling multi-label classification.	Transformers require high computational resources. Models struggle with imbalanced datasets. Financial texts present unique language challenges.	MarIA and BETO excel with high macro F1-scores. Target detection accuracy reached 69.74% for MET.
6	A Comparison Of Lexicon-Based And ML-Based Sentiment Analysis: Are There Outlier Words?	Siddhant Jaydeep Mahajani, Shashank Srivastava, Alan F. Smeaton	Compared Hedonometer , Microsoft Azure sentiment analysis across domains.	Identifies domain-specific differences in sentiment analysis. Highlights strengths and weaknesses of Hedonometer.	Limited by smaller Hedonometer lexicon subset. Linear regression may miss sentiment complexities.	No major outliers found; Hedonometer adjustments likely minimal.
7	A Comparative Study Of Lexicon-Based And Machine Learning-Based Classifications In Sentiment Analysis	T. Nikil Prakash, A. Aloysius	Comparative analysis of lexicon-based and machine learning methods, including SVM, Naive Bayes, and KNN.	Comprehensive comparison of sentiment analysis approaches. Highlights accuracy and performance differences	Study limitations include specific datasets and algorithms. Performance varies by text	Machine learning methods offer better accuracy and performance. Lexicon-based methods are

				between methods.	domain and features.	simpler but less adaptable.
8	Text Analysis In Finance: The Challenges For Efficient Application	Karol M. Klimczak	Overview and analysis of text analysis methods in finance, including sentiment analysis, topic identification, disambiguation, and multilingual processing.	Highlights the potential of text analysis in improving financial decision-making. Identifies domain-specific tools and methods tailored to the financial industry.	Current text analysis methods show low accuracy. Financial texts have unique characteristics complicating tool application.	Text analysis can improve financial forecasting, client management, early-warning systems, and risk analysis, but requires better accuracy and method adaptation.
9	Comprehensive Review Of Text-Mining Applications In Finance	Aaryan Gupta, Vinya Dengre, Hamza Abubakar Kheruwala, Manan Shah	Text mining with sentiment analysis, NLP, deep learning.	Wide applications in forecasting, banking, and finance. Automates and improves accuracy in analyzing financial text. Combines traditional methods with deep learning for advanced predictions.	Relies on domain-specific datasets and complex models. Requires extensive preprocessing Some models struggle with generalization across financial contexts..	Deep learning and NLP's accuracy, with models achieving 83% accuracy. Sentiment analysis aids in predicting stock prices and trends. Deep learning models outperform traditional ones in unstructured financial data.
10	Behaviour And Markets: The Interaction Between Sentiment Analysis And Ethical Values	Jason A. Cook, Khurshid Ahmad	Sentiment Analysis, Bag-of-Words, Autoregression.	Identified correlations between moral language and sentiment. Used large datasets for quantitative analysis of sentiment's market impact.	Results may vary with moral term set; expansion needed. Moral terms had less impact on return variability compared to negative sentiment.	Negative and vice terms showed 46.30% correlation; positive and virtue terms showed 30.38% correlation. Negative sentiment terms improved autoregressive model performance, while vice terms did not.
11	Sentiment Analysis Of Financial News:	Argimiro Arratia, Gustavo	Textual Corpus Management,	Widely used in finance for constructing sentiment	Focuses on lexicon-based methods, possibly	Constructed six sentiment indicators;

	Mechanics And Statistics	Avalos, Alejandra Cabaña, Ariel Duarte-López Martí Renedo-Mirambell	Computing Sentiment Scores, Aggregating Sentiment Scores, Modelling	indicators. Offers insights into statistical properties and predictive power.	overlooking machine learning approaches. Lexicon accuracy depends on the lexicon's quality.	analyzed Dow Jones news. Found strong correlation between negative sentiment and market volatility. News volume surged around earnings announcements.
12	Sentiment Analysis In Financial Texts	Samuel W. K. Chan , Mickey W. C. Chong	SAE uses machine learning for sentiment analysis, leveraging POS tags, PMI, LR, and TTFs. Builds parse trees and uses SentiWordNet and LIWC for sentiment assessment.	Language-independent Efficient Processing Contextual Understanding Addresses Cognitive Limitations	Financial Text Nuances Dependence on Lexicons	High Parsing Accuracy Effective Sentiment Analysis Statistically Significant Findings
13	Sentiment Analysis Using Domain-Adaptation And Sentence-Based Analysis	Gizem Gezici Berrin Yanikoglu Dilek Tapucu Yücel Saygın	Domain Adaptation Sentence-Based Analysis Feature Integration	Domain Specificity Sentence-Level Insights Simplicity and Speed	Information Loss from Pruning Limited Impact of Seed Words	State-of-the-art Performance Impact of Sentence Features Challenges of Multi-Class Classification
14	Domain Adaptive Learning For Multi Realm Sentiment Classification On Big Data	Maha Ijaz Naveed Anwar Mejdl Safran Sultan Alfarhood Tariq Sadad Imran	Proposes Transfer Learning-based MDSC using pre-trained models and deep learning. Compares BERT, RoBERTa, ELECTRA, ULMFiT, CNN, and LSTM.	Efficiency and Reusability Improved Performance Domain Adaptability	Dependence on Pre-trained Models Computational Requirements Challenges with Multi-Class Sentiment Classification	Superiority of Transfer Learning Impact of Dataset Size Effectiveness of FARM Framework Performance Variations

			Involves data cleaning and preprocessing.			Across Domains
15	Sentiment Analysis For Effective Stock Market Prediction	Shri Bharathi Angelina Geetha	Combines sentiment analysis with moving averages for stock prediction. Uses NLP and SSS algorithm for analysis.	Improved Accuracy Real-Time Insights Public Sentiment Integration	Single Company and Stock Exchange Focus Dependence on RSS Feed Quality and Availability Simplification of Sentiment Analysis	Enhanced Prediction Accuracy Positive Impact of Sentiment Integration Potential for Further Improvement
16	Stock Trend Prediction Using Sentiment Analysis	Qianyi Xiao Baha Ihnaini	Uses VADER, FinBERT, and machine learning classifiers.	Integration of Multiple Data Sources Time Sensitivity Holiday Effect Adjustment Comparative Analysis	Focus on Short-Term Predictions Limited Stock Selection Dependence on Data Quality	Naïve Bayes as the Best Classifier Effectiveness of Opening Hours Division Improved Performance with Combined Sentiment and Holiday Adjustment
17	Impact Of Social Media Sentiments In Stock Market Predictions: A Bibliometric Analysis	Deepshi Garg Prakash Tiwari	Conducted bibliometric analysis of 1,450 documents on social media sentiment in stock prediction. Used VOSviewer for network maps.	Comprehensive Overview Identification of Key Contributors Research Gap Identification	Database Bias Language Limitation Timeframe Constraint	Research Growth Key Journals and Authors Influential Countries and Institutions Co-authorship Networks Keyword Trends
18	Sentiment Analysis Of Financial News Using	Anita Yadava C K Jhaa	Explores financial news sentiment analysis using	Unsupervised Learning	Limited Dataset	Noun-Verb Approach Performs Best

	Unsupervised Approach	Aditi Sharanb Vikrant Vaishb	semantic orientation. Enhances Turney's PMI-IR with market-specific indicators and a seed set approach. Evaluates pattern, hybrid, and noun-verb methods.	Domain-Specific Sentiment Indicators Improved Feature Selection	Manual Intervention Complexity of Financial Language	Hybrid and Noun-Verb Outperform Baseline Turney's Approach Shows Higher Recall
19	A Mass Psychological Perspective On Financial Markets	Thomas Fenzl Thomas Brudermann Constantin Malik Linda Pelzmann	Explores financial market cycles using behavioral and social psychology insights. Focuses on 2008-2009 crisis.	Interdisciplinary Approach Focus on Social Dynamics Long-Term Perspective	Theoretical Framework Limited Predictive Power Focus on Western Markets	First and Second Order Phenomena Culture of Indebtedness Boom-Bust Cycles as Second-Order Phenomena
20	Challenges In Sentiment Analysis	Saif M. Mohammad	Comprehensive overview of sentiment analysis challenges. Covers tasks, lexicons, composition, annotation, multilingual aspects, and applications.	Comprehensive Scope In-Depth Analysis Current State of Research Future Directions	Theoretical Focus Rapidly Evolving Field	Sentiment analysis tasks, lexicon limitations, composition complexity, annotation challenges, multilingual issues.

III. Conclusion :

The Financial sector has seen a hike in the usage of the Sentiment analysis tool, which not only helps in market forecasting but also gives a strong hand in managing the financial risks as well as making investment decisions. This review presents the movement from the early lexicon-based methods to more sophisticated machine-learning approaches for sentiment analysis. Initially, the lexicon-based techniques were simpler and better-performing, but the machine learning methods were able to deliver much more precise and adaptable responses, particularly with the introduction of transformer networks. The appearance of hybrid models that bring the best part of both technologies opens the way for unprecedented research. However, some issues such as domain-specific lexicons, context-sensitive analysis, and the complexity related to financial language management remain untouched. The engineers of the field will play an essential part in achieving these aims and thus bridging the gap between the requirement and the opportunity of sentiment analysis in the finance sector.

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